

FUNCTION

In mathematics, a function is a relation between input x and output y sets, where each x maps to a unique y , denoted as $y = f(x)$.

This notation means that for a given 'x' value, the function 'f' will produce a 'y' value.

We have conducted a study and found a linear relationship between the number of cigarettes smoked daily and the risk of developing lung cancer over a 40-year span.

- If a person does not smoke, the risk is **1%**.
- For every cigarette smoked daily, the risk increases by **0.5%**.

$$y = 0.01 + 0.005x$$

↓
Lung cancer risk
over 40 years

↓
The # of cigarettes
smoked daily



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$$y = 0.01 + 0.005 (\text{🚬🚬})$$

This individual's
40-year lung cancer
risk is 2%

